

13 THERMODYNAMIC SYSTEMS

H2 Physics 9478

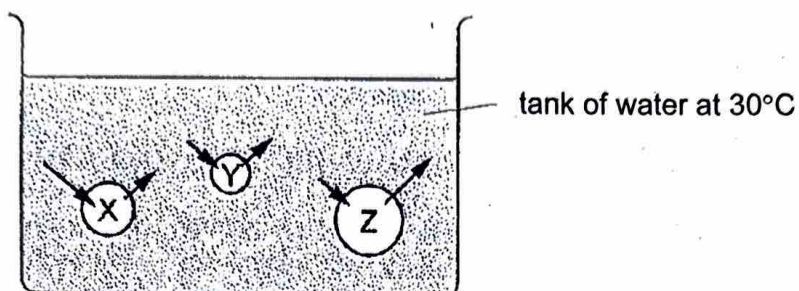


13.1 Self-Check Questions

- S1** Explain what is meant by the internal energy U of
- (i) a system,
 - (ii) an ideal gas.
- S2** State how the thermodynamic temperature of an ideal gas is related to the average microscopic kinetic energy of the molecules of the gas.
- S3** Two bodies are in thermal equilibrium. What does this statement mean?
- S4** With reference to heat transfer, explain what is meant by
- (i) two bodies having the same temperature,
 - (ii) body H having a higher temperature than body C.
- S5** Explain how work done by a gas is positive during expansion under constant pressure.
- S6** State the zeroth law of thermodynamics and explain its importance in relation to the use of a thermometer to measure temperature.
- S7** State the first law of thermodynamics and explain how it is an application of the principle of conservation of energy.
- S8** Explain the meaning of the terms *isobaric*, *isovolumetric*, *isothermal* and *adiabatic* when used to describe thermodynamic processes.
- S9** Define specific heat capacity, and outline the main principles of its determination for
- (i) a solid,
 - (ii) a liquid.
- S10** Define specific latent heat, and outline the main principles of its determination for
- (i) fusion,
 - (ii) vaporisation.
- S11** Explain, using the kinetic model of matter, why
- (i) melting and boiling take place without a change in temperature.
 - (ii) the specific latent heat of vaporisation is higher than the specific latent heat of fusion for the same substance,
 - (iii) cooling effect accompanies evaporation.

13.2 Self-Practice Questions

- SP1** Three objects X, Y and Z are put into a tank of water kept at 30°C . The arrows represent the input and the output of thermal energy for each object just after being placed in the tank. The length of each arrow corresponds to the quantity of energy transferred.

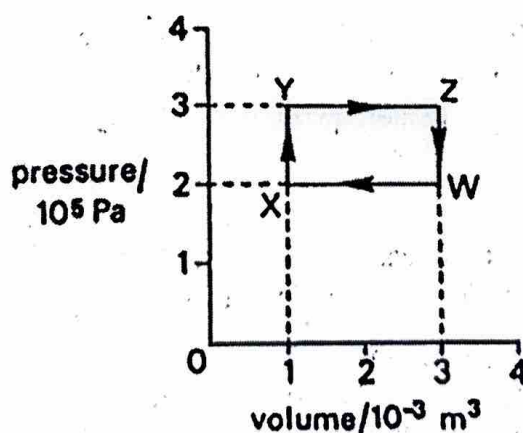


What are the initial temperatures of X, Y and Z?

	X	Y	Z
A	Above 30°C	At 30°C	Below 30°C
B	Above 30°C	Below 30°C	Above 30°C
C	Below 30°C	Above 30°C	Below 30°C
D	Below 30°C	At 30°C	Above 30°C

[N18/1/13]

- SP2** A gas undergoes the cycle of pressure and volume changes $W \rightarrow X \rightarrow Y \rightarrow Z \rightarrow W$ shown in the diagram.



What is the net work done by the gas?

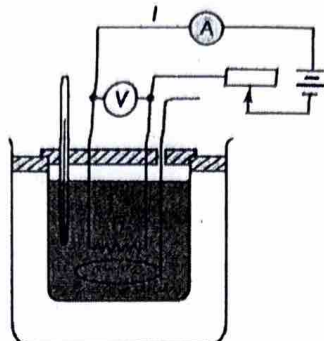
- | | |
|-----------------|-----------------|
| A -600 J | B -200 J |
| C 0 J | D 200 J |
| E 600 J | |

[N89/1/3]

**RAFFLES INSTITUTION
YEAR 56 PHYSICS DEPARTMENT**

- SP3** A 1.00 mol sample of argon gas is heated at constant pressure from 300 K to 420 K. Assuming that the gas is ideal, calculate
- (a) the work done on the gas,
 - (b) the increase in its internal energy and
 - (c) the energy transferred to the gas by heat.
- SP4** An ideal gas contained within a piston-cylinder assembly is compressed. 500 J of work is done and 200 J of heat is lost. Calculate the increase in the internal energy of the gas.
- SP5**
- (a) Explain what is meant by an isothermal change.
 - (b) How is an isothermal compression of a gas achieved in practice?
 - (c) Explain why, in the isothermal compression of an ideal gas, the internal energy remains the same even though mechanical work is done on the system.
- SP6** A cylinder fitted with a frictionless piston contains $5.0 \times 10^{-4} \text{ m}^3$ of an ideal gas at a pressure of $1.0 \times 10^5 \text{ Pa}$ and temperature of 300 K.
- The gas is then
- (i) heated at constant pressure to 450 K, and then
 - (ii) cooled at constant volume to the original temperature of 300 K. The heat extracted in this stage is 37.5 J.
- (a) Illustrate these changes on a p-V diagram labelled with the appropriate values of pressure and volume.
 - (b) Determine the work done by the gas in stage (i).
 - (c) Calculate the total heat input in stage (i).
- SP7** A thermally insulated tube through which a gas may be passed through at constant pressure contains an electric heater and thermometers for measuring the temperature of the gas as it enters and as it leaves the tube. $3.0 \times 10^{-3} \text{ m}^3$ of the gas of density 1.8 kg m^{-3} flows into the tube in 90 s and, when electrical power is supplied to the heater at a rate of 0.16 W, the temperature difference between the outlet and inlet is 2.5 K.
- (a) Neglecting heat loss to the surroundings, calculate the specific heat capacity of the gas at constant pressure.
 - (b) Explain the features of the constant-flow method that makes it particularly suitable for the accurate determination of the specific heat capacities of fluids.

- SP8** The figure shows the apparatus used in the determination of the specific heat capacity of a liquid. The total heat capacity of the calorimeter, the heater, the stirrer and the thermometer was 107 J K^{-1} . The mass of liquid in the calorimeter was 0.241 kg , and the ammeter and voltmeter readings were 3.40 A and 12.2 V respectively. When the temperature of the liquid was equal to that of the surroundings, its value was found to be increasing at the rate of $46.3 \times 10^{-3} \text{ K s}^{-1}$.



- (a) Determine a value for the specific heat capacity of the liquid.

State the advantage of measuring the rate of rise of temperature of the liquid when its

- (b) temperature is equal to that of the surroundings rather than measuring the rise of temperature taking place in a given length of time.

- SP9** A well-lagged copper calorimeter of mass 120 g contains 200 g of water and 50 g of ice, initially at 0°C . Steam at 100°C is blown through the water until the temperature reaches 30°C . Calculate the mass of the steam that must have condensed.

Specific heat capacity of copper = $380 \text{ J kg}^{-1} \text{ K}^{-1}$

Specific latent heat of vaporisation of water = $2.26 \times 10^6 \text{ J kg}^{-1}$

- SP10** An immersion heater takes time T_1 to raise the temperature of a mass M of a liquid from a temperature t_1 to its normal boiling point t_2 . In a further time T_2 , a mass m of the liquid is vaporised. If the specific heat capacity of the liquid is c and heat losses to the atmosphere and to the containing vessel are ignored, the specific latent heat of vaporisation is

A $\frac{Mc(t_2 - t_1)T_2}{mT_1}$

B $\frac{mc(t_2 - t_1)T_2}{MT_1}$

C $\frac{Mct_1T_2}{mT_1}$

D $\frac{mT_1}{Mc(t_2 - t_1)T_2}$

E $\frac{mT_1}{Mct_1T_2}$

[N77/2/29]

**RAFFLES INSTITUTION
YEAR 56 PHYSICS DEPARTMENT**

SP11 In a heating experiment, energy is supplied at a constant rate to a liquid in a beaker of negligible heat capacity. The temperature of the liquid rises at 4.0 K per minutes just before it begins to boil. After 40 minutes all the liquid has boiled away.

For this liquid, what is the ratio $\frac{\text{specific heat capacity}}{\text{specific latent heat of vaporisation}}$?

A $\frac{1}{10} \text{ K}^{-1}$

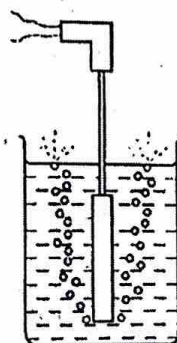
B $\frac{1}{40} \text{ K}^{-1}$

C $\frac{1}{160} \text{ K}^{-1}$

D $\frac{1}{640} \text{ K}^{-1}$

[J00/1/25]

SP12 In an experiment to find its specific latent heat of vaporisation, water is vaporised using an immersion heater as shown.



Two sources of error in this experiment are:

Error 1 water splashing out of the container;

Error 2 vapour condensing on the handle of the heater and dripping back into the container

What is the effect of these two experimental errors on the calculated value for the specific latent heat?

	Error 1	Error 2
A	Decrease	Decrease
B	Decrease	Increase
C	Increase	Decrease
D	Increase	Increase

[J99/1/24]

SP13 A constant power supply is used to melt 1 kg of ice, to heat the water produced, and finally to turn all the water to steam.

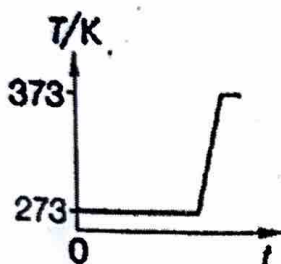
Specific heat capacity of water $4 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$

Specific latent heat of fusion of ice $3 \times 10^5 \text{ J kg}^{-1}$

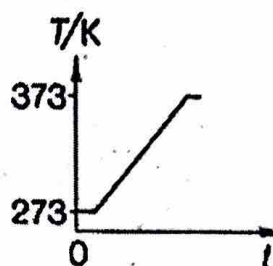
Specific latent heat of vaporisation of water $2 \times 10^6 \text{ J kg}^{-1}$

Which graph best shows how the thermodynamic temperature T varies with time t for this sequence?

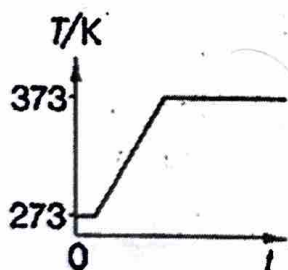
A



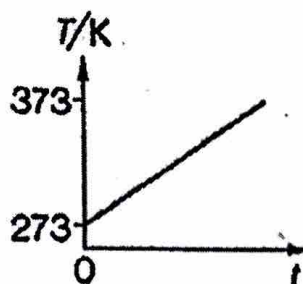
B



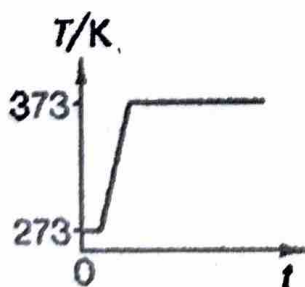
C



D



E



[J91/1/25]

**RAFFLES INSTITUTION
YEAR 56 PHYSICS DEPARTMENT**

13.3 Discussion Questions

❖ **Internal energy**

- D1** (a) (i) A car tyre has a fixed internal volume of 0.0120 m^3 . On a day when the temperature is 25°C , the pressure in the tyre has to be increased from $2.62 \times 10^5 \text{ Pa}$ to $3.23 \times 10^5 \text{ Pa}$. Assuming the air is an ideal gas, calculate the amount of air in moles which has to be supplied at constant temperature. [3]
- (ii) A portable supply of air used to inflate tyres has a volume of 0.0108 m^3 and is filled with air at a pressure of $8.72 \times 10^5 \text{ Pa}$. Show that at 25°C , there is more than enough air in it to supply four tyres as in (i), without the pressure falling below $3.23 \times 10^5 \text{ Pa}$. [2]
- (b) (i) Show that the internal energy of a molecule of air at a temperature of 25°C is $6.17 \times 10^{-21} \text{ J}$. Assume the air behaves as an ideal gas. [2]
- (ii) Hence calculate the internal energy of one mole of air at a temperature of 25°C . [1]
- (iii) Calculate the increase in the internal energy of the air in the tyre in (a)(i) as a result of increasing its pressure. [2]

[N2007/P3/6(part)]

- D2** In a space, such as a swimming pool enclosure, water at 30°C and water vapour, also at 30°C coexist.
- (a) Compare the pattern of movement and the speed of molecules in water and water vapour at the same temperature. [4]
- (b) State what is meant by the *internal energy* of a system. [2]
- (c) Using your answer in (b), compare the internal energy per unit mass of water and water vapour at the same temperature. [3]
- (d) Explain, in terms of internal energy, why the specific latent heat of vaporisation of a substance is greater than its specific latent heat of fusion. [2]

[N2006/P3/2(part)]

❖ **Heating and work done**

- D3** (a) State what is meant by the thermal equilibrium of two bodies by reference to
- (i) temperature, [3]
- (ii) movement of thermal energy. [2]
- (b) A student carries out an experiment with melting ice, using the apparatus illustrated in Fig. 3.1.

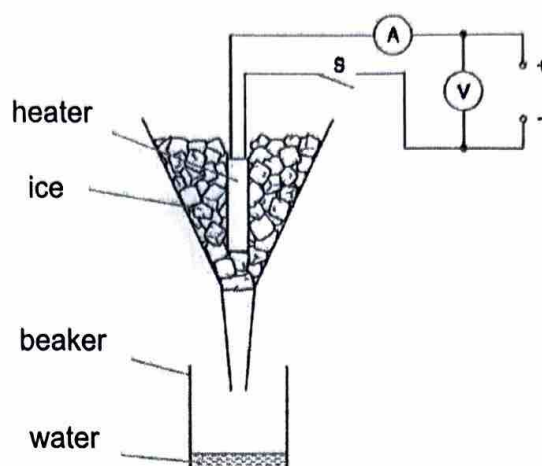


Fig. 3.1

Initially, the switch S is open, and the funnel is kept filled with pure melting ice.

Water drips out of the funnel at a constant rate.

(i) Suggest and explain whether the ice is in thermal equilibrium with its surroundings. [1]

(ii) The switch S is closed. When water drips out of the funnel at a different constant rate, the beaker is replaced with an empty beaker and the readings shown in Fig. 3.2 are obtained.

Ammeter reading / A	Voltmeter reading / V	Mass of empty beaker / g	Mass of beaker plus water / g	Time of collection of water / minutes
6.3	12.0	32.4	114.0	5.0

Fig. 3.2

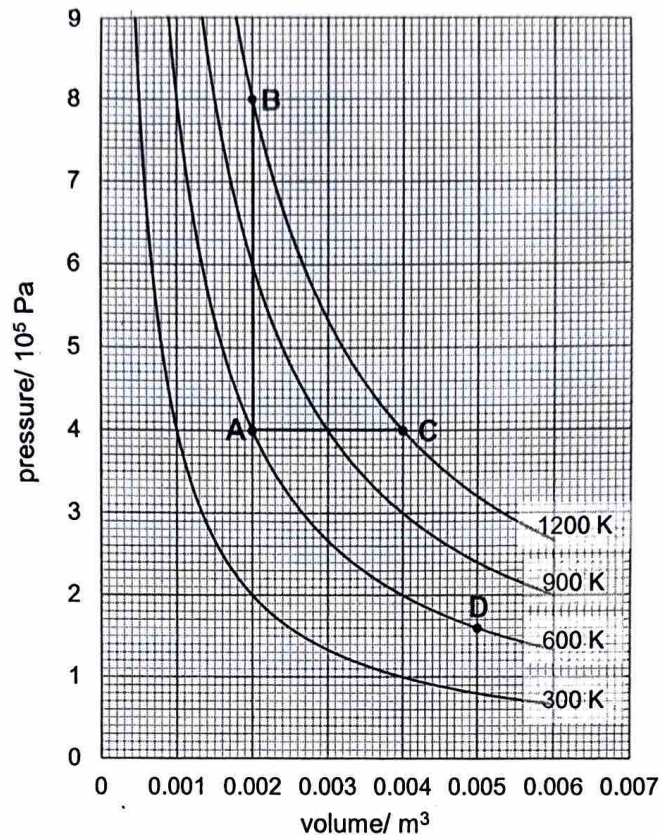
The specific latent heat of fusion of ice is 330 J g^{-1} .

Calculate the rate R at which thermal energy is gained from the atmosphere by the ice in the funnel. [3]

[N2017/3/8(a), (b)]

RAFFLES INSTITUTION
YEAR 56 PHYSICS DEPARTMENT

- D4** The figure below gives data in graphical form concerning the pressure, volume and temperature of a fixed mass of gas. On each curve, the pressure is plotted against the volume at a fixed temperature.



- (a) Write down the ideal gas equation and use it to find the number of moles of gas. [3]
- (b) Show that the data of the graph are consistent with the gas behaving as an ideal gas over the range of temperature from 300 K to 1200 K. [3]
- (c) The gas has a heat capacity of 3.33 J K^{-1} at a constant volume and 4.66 J K^{-1} at constant pressure.
Using the values provided, calculate the quantity of heat required to take the gas
 - (i) from A to B along the line AB.
 - (ii) from A to C along the line AC. [3]
- (d) The internal energy of the gas at A is 2000 J. Find the internal energy at B and at C. [3]
- (e) How much work is done by the gas if an expansion at constant pressure takes place from A to C? [2]
- (f) Explain why internal energy of the gas at D is also 2000 J. [2]

[J88/P2/Q10(part)]

❖ **Laws of thermodynamics**

D5 An adiabatic change is one in which no heat is supplied or extracted.

- (a) Write down an expression for the first law of thermodynamics when related to an adiabatic change in the state of an ideal gas in which the gas does 600 J of work.
- (b) Deduce the change in internal energy of the gas.
- (c) What can you deduce about the temperature of the gas? [3]

[N89/P2/Q4]

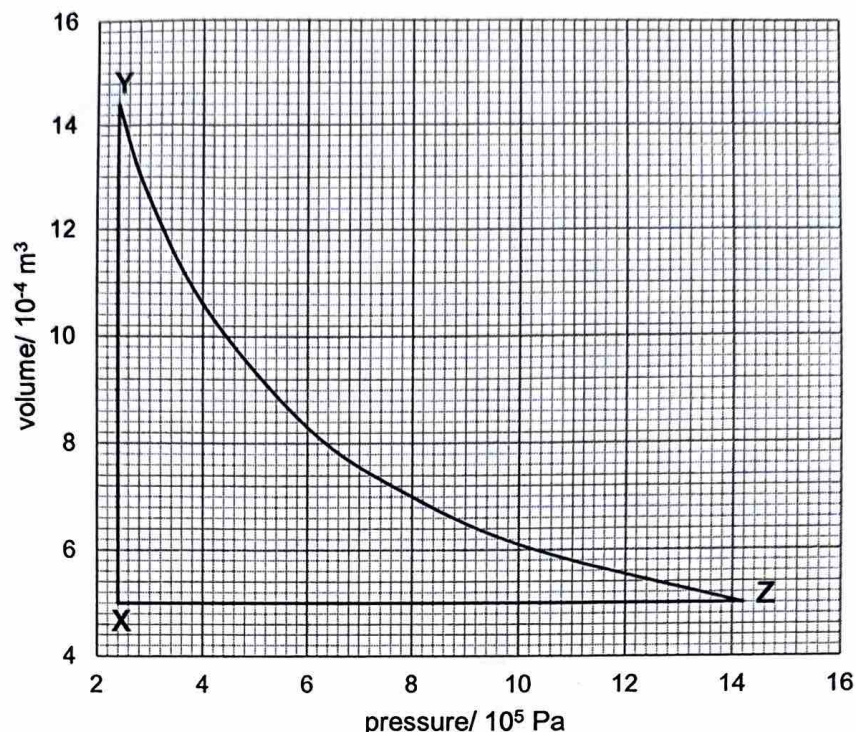
D6 A mass of 5.6 g of nitrogen gas occupies a volume of $4.6 \times 10^3 \text{ cm}^3$ at a pressure of $1.0 \times 10^5 \text{ Pa}$ and a temperature of 280 K. The specific heat capacity of nitrogen gas, when heated at constant volume, is $0.73 \text{ J g}^{-1} \text{ K}^{-1}$. Nitrogen may be assumed to be an ideal gas.

- (a) Calculate the *thermal energy* required to raise the temperature of the gas by 1 K at constant volume. [2]
- (b) The temperature of the gas is raised from 280 K to 281 K at constant pressure. Determine, for this temperature change,
 - (i) the change in volume of the gas, [2]
 - (ii) the external work done by the gas. [2]
- (c) (i) State the first law of thermodynamics, indicating the directions of all energy changes. [2]
- (ii) Use the first law and your answers in (a) and (b) to determine the *thermal energy* required to raise the temperature of the nitrogen gas from 280 K to 281 K when the gas is heated at constant pressure. [1]

[N2011/P3/Q2]

RAFFLES INSTITUTION
YEAR 56 PHYSICS DEPARTMENT

D7 A fixed mass of an ideal gas undergoes the cycle of changes XYZX as shown in the figure.



At point X, the temperature of the gas is 290 K.

- (a) Calculate the amount of gas. [2]
- (b) Determine, for this mass of gas, the magnitude of
- (i) the work done during the change from X to Y, [2]
- (ii) the change in the internal energy during one complete cycle XYZX. [1]
- (c) Some energy changes during one cycle XYZX are shown in the following table.

change	work done on gas / J	heat supplied to gas / J	increase in internal energy / J
X → Y		+570	
Y → Z	+ 540	0	
Z → X	0		

Use your answers in (b) to complete the table.

[3]

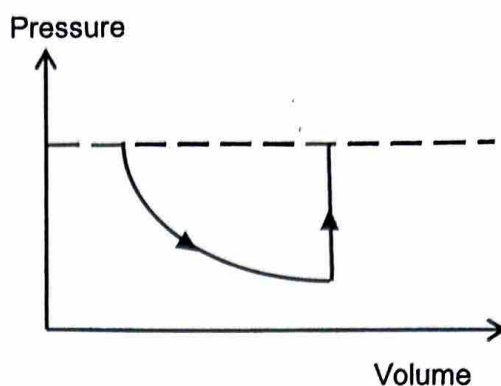
[N2014/P3/Q3]

- D8 (a)** A sample of an ideal gas initially having internal energy U_1 is allowed to expand adiabatically by doing external work W .

Heat Q is then supplied to it, keeping the volume constant at its new value, until the pressure rises to its original value.

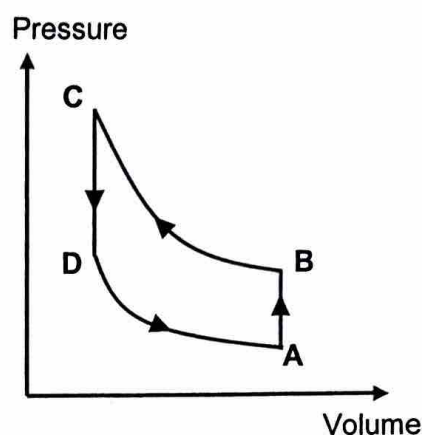
The internal energy is then U_2 .

Determine the total increase in internal energy in terms of Q and W . [2]



- (b)** An ideal gas undergoes the following cyclic process ABCDA as shown. Use the first law of thermodynamics to complete the following table.

Process	Q/J	W/J	$\Delta U/J$
AB	280		
BC	20	170	
CD	-400		
DA		-70	



❖ Specific heat capacity and specific latent heat

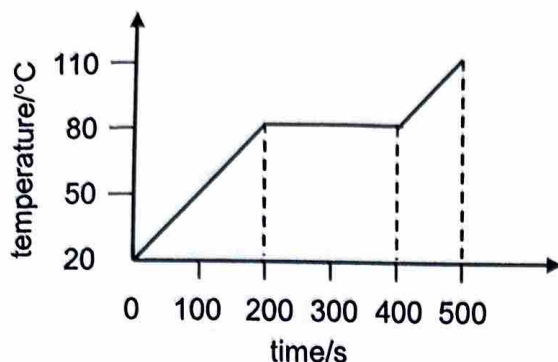
D9 Data:

Specific heat capacity of water	$4190 \text{ J kg}^{-1} \text{ K}^{-1}$
Specific latent heat of fusion of water	$3.33 \times 10^5 \text{ J kg}^{-1}$
Specific latent heat of vaporisation of water	$2.26 \times 10^6 \text{ J kg}^{-1}$
Specific heat capacity of ice	$2050 \text{ J kg}^{-1} \text{ K}^{-1}$
Molar gas constant, R	$8.31 \text{ J K}^{-1} \text{ mol}^{-1}$
Boltzmann constant, k	$1.38 \times 10^{-23} \text{ J K}^{-1}$
Avogadro constant, N_A	$6.02 \times 10^{23} \text{ mol}^{-1}$

A thermally isolated beaker contains 1.000 kg of water at 20°C . A mass M of ice at -20°C is dropped into it. When the system reaches thermal equilibrium, a mass m of ice remains unmelted. Show that $m = 1.123M - 0.252$, where m and M are measured in kg. (Assume that the heat transferred to the beaker is negligible.)

**RAFFLES INSTITUTION
YEAR 56 PHYSICS DEPARTMENT**

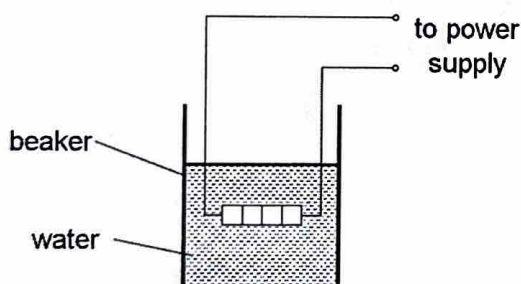
- D10** A certain solid X is heated in an insulated container at a constant rate. The temperature of the material varies as shown in the graph. If the specific heat capacity of the solid is $1800 \text{ J kg}^{-1} \text{ K}^{-1}$, calculate



- (a) the specific latent heat of fusion of the solid,
- (b) the specific heat capacity of liquid X.

- D11** A student carries out an experiment to determine the specific latent heat of vaporisation of water. Water is boiled in a beaker by means of an electric heater, as shown in the figure. The power supplied to the heater is measured when the water is boiling at a constant rate. The mass m of water evaporated in 5.0 minutes is determined.

Data for the power P and the mass m for two different values of P are shown.



P / W	m / g
140	14.1
95	8.2

- (a) Suggest why, in order to obtain a reliable result for the specific latent heat, the mass m is determined for two different values of P . [1]
- (b) Calculate the value for the specific latent heat of vaporisation. [3]

[N2010/P3/1(part)]

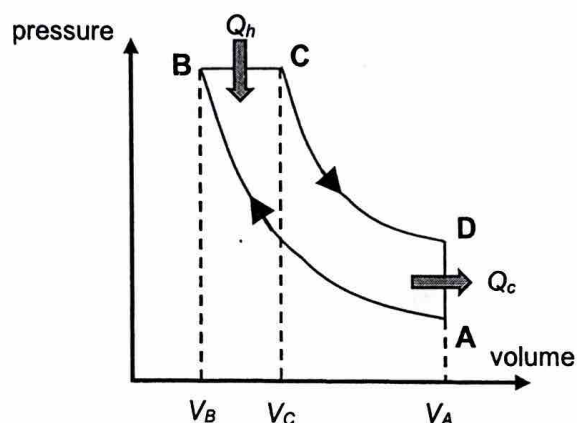
13.4 Challenging Questions

C1 A multicylinder gasoline engine in an airplane, operating at $2500 \text{ rev min}^{-1}$, takes in energy $7.89 \times 10^3 \text{ J}$ and exhaust $4.58 \times 10^3 \text{ J}$ for each revolution of the crankshaft.

- How many litres (L) of fuel does it consume in 1 hour of operation if the heat of combustion is $4.03 \times 10^7 \text{ J L}^{-1}$?
- What is the mechanical power output of the engine? Ignore friction.
- What power must the exhaust and cooling system transfer out of the engine?

C2 An idealised diesel engine operates in a cycle known as the air-standard diesel cycle as shown below. Fuel is sprayed into the cylinder at the point of maximum compression, B. Combustion occurs during the expansion $B \rightarrow C$, which is modeled as an isobaric process.

Processes $A \rightarrow B$ and $C \rightarrow D$ are adiabatic processes.



Show that the efficiency of an engine operating in this idealized diesel cycle is

$$e = 1 - \frac{1}{\gamma} \left(\frac{T_D - T_A}{T_C - T_B} \right)$$

Additional information given for C2:

- The *principal molar heat capacity* of a gas at *constant pressure*, C_P , is the amount of heat needed to raise the temperature of one mole of the gas by 1 K at constant pressure.
- The *principal molar heat capacity* of a gas at *constant volume*, C_V , is the amount of heat needed to raise the temperature of one mole of the gas by 1 K at constant volume.
- For an adiabatic change (or process), $pV^\gamma = \text{constant}$, where $\gamma = \frac{C_P}{C_V}$
- $C_P - C_V = R$, the ideal gas constant.

The efficiency of a heat engine is defined as $e = \frac{\text{net work done in 1 cycle}}{\text{heat supplied in 1 cycle}}$

**RAFFLES INSTITUTION
YEAR 56 PHYSICS DEPARTMENT**

13.5 Numerical Answers

- D1** (a) (i) 0.295 mol
(b) (ii) 3710 J
(b) (iii) 1090 J
- D3** (b) (ii) 14.2 W
- D4** (a) 0.16 mol
(c) (i) 2000 J
(ii) 2800 J
(d) 4000J
(e) 800 J
- D6** (a) 4.09 J
(b) (i) 16 cm³
(ii) 1.6 J
(c) (ii) 5.69 J
- D7** (a) 0.050 mol
(b) (i) 230 J
(ii) 0 J
(c) -230J, +340 J, +540 J, -880 J, -880 J
- D8** (b) 0 J, 280 J, 190 J, 0 J, -400 J, 0 J, -70 J
- D10** (a) $1.08 \times 10^5 \text{ J kg}^{-1}$
(b) $1800 \text{ J kg}^{-1} \text{ K}^{-1}$
- D11** (b) $2.3 \times 10^6 \text{ J kg}^{-1}$
- C1** (a) 29.4 L
(b) $1.38 \times 10^5 \text{ W}$
(c) $1.91 \times 10^5 \text{ W}$

13.6 Self-Check & Self-Practice Suggested Solutions

- S1 (i) The internal energy of a system is the sum of the microscopic kinetic energy, due to the random motion of the molecules, and the microscopic potential energy, associated with the intermolecular forces of the system.
- (ii) For an ideal gas, the assumption is that the molecules have negligible forces of interactions with each other. Thus, the microscopic potential energy associated with ideal gas molecules is zero. The internal energy of an ideal gas is therefore the sum of all the microscopic kinetic energies of the gas molecules.
- S2 The average microscopic kinetic energy of a molecule of the gas is proportional to the thermodynamic temperature T of the gas: $\langle E_k \rangle = \frac{3}{2} kT$
- S3 It means that there is no net transfer of heat between the two bodies and the two bodies are at the same temperature.
- S4 (i) There is no net heat transfer between the two bodies.
- (ii) If body H and body C is in thermal contact, then there will be heat transfer from body H to body C.
- S5 Refer to notes pg 9 to 10.
- S6 Zeroth Law of Thermodynamics states that if objects A and B are each separately in thermal equilibrium with a third object C, then A and B are in thermal equilibrium with each other.
- Zeroth Law of Thermodynamics explains the basis for temperature measurement. The comparison of temperatures of A and B can be done with C without making direct thermal contact between A and B i.e. C acts as a thermometer.
- S7 The first law of thermodynamics states that the increase in the internal energy of a system is equal to the sum of the heat supplied to the system and the work done on the system, and the internal energy of a system depends only on its state.
- The principle of conservation of energy states that energy cannot be created or destroyed. It can only be transferred from one store to another.
- Since the internal energy of a system is the sum of the microscopic kinetic energy and microscopic potential energy associated with the molecules of the system, it cannot increase unless there are other forms of energy entering the system.
 - In a thermodynamic system, the energy that enters a system can be in the form of heat supplied and/or work done on the system. These are transformed into internal energy of the system, thereby increasing internal energy.
- S8 An *isobaric* process is one where the system undergoes heat exchange such that the volume change occurs at constant pressure.

An *isovolumetric* process is one where the system undergoes heat exchange such that the pressure change occurs at constant volume.

An *isothermal* process is one where changes in pressure and volume of a system take place without any change in temperature. This can be achieved by allowing the (heat exchange) process to occur slowly in a thin-walled vessel and by surrounding the system in a constant temperature bath.

**RAFFLES INSTITUTION
YEAR 56 PHYSICS DEPARTMENT**

An *adiabatic* process is one where the pressure and volume of the system change without any heat exchange. This can be achieved if the process occurs rapidly and if the system is insulated.

S9 Refer to notes Pg 28 to 31.

S10 Refer to notes Pg 33 to 36.

S11 Refer to notes Pg 25 to 27.

SP1 X: $Q_{\text{net}} < 0$ i.e. $T_{x,\text{initial}} < 30^\circ\text{C}$

Y: $Q_{\text{net}} = 0$ i.e. Y is at thermal equilibrium with its surrounding $T_{y,\text{initial}} = 30^\circ\text{C}$

Z: $Q_{\text{net}} > 0$ i.e. $T_{z,\text{initial}} > 30^\circ\text{C}$

Answer: D

SP2 Net work done by the gas = area enclosed by the rectangle WXYZ
 $= (1 \times 10^5) (2 \times 10^{-3}) = +200 \text{ J}$

Answer: D

SP3 (a) $W = -p(\Delta V) = -nR(\Delta T)$
 $= -1.00(8.31)(420 - 300) = -997 \text{ J}$

(b) $\Delta U = \frac{3}{2}nR(T_2 - T_1)$
 $= \frac{3}{2}(1.00)(8.31)(420 - 300)$
 $= 1495.8 \text{ J} \approx 1500 \text{ J}$

(c) First law of thermodynamics,

$$\Delta U = Q + W$$

$$Q = \Delta U - W = 1495.8 - (-997.2) = 2493 = 2490 \text{ J} \quad (3\text{sf})$$

Heat energy transferred to the gas is 2490 J.

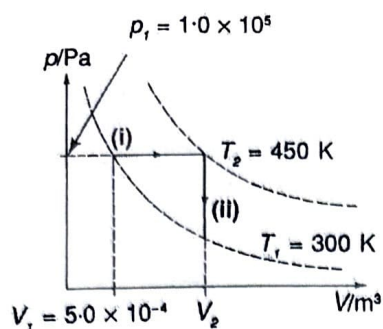
SP4 $\Delta U = Q + W = (-200) + 500 = 300 \text{ J}$

SP5 (a) Isothermal change refers to the change in pressure and volume of a gas under a constant temperature condition, i.e. $\Delta T = 0$.

(b) Isothermal compression of a gas can be achieved in practice by keeping the gas in a thin-walled, and good conducting vessel placed in a temperature bath. The compression is done slowly, in small steps, to allow for heat flow out of the gas to maintain a constant temperature.

(c) For isothermal compression, the rate of work done on the gas is the same as the rate at which heat is extracted from the gas; hence there is no change in the internal energy.

SP6 (a)



(b) $W = p_1 \Delta V = p_1 (V_2 - V_1)$

Using $\frac{V_1}{T_1} = \frac{V_2}{T_2}$ at constant p ,

$$V_2 = \frac{T_2}{T_1} V_1 = \frac{450}{300} \times V_1$$

$$\begin{aligned} W &= p_1 (V_2 - V_1) \\ &= (1.0 \times 10^5) \left(\frac{450}{300} V_1 - V_1 \right) \\ &= (1.0 \times 10^5) \left(\frac{450}{300} - 1 \right) (5.0 \times 10^{-4}) \\ &= 25 \text{ J} \end{aligned}$$

- (c) From stage (ii), for a constant volume process, applying the first law of thermodynamics:
 $\Delta U' = Q' + W' = Q' = -37.5 + 0 = -37.5 \text{ J}$

Since ΔU only depends on ΔT for an ideal gas, there is no change in ΔU at the end of stage (ii) since the final temperature is the same as the initial temperature.

For stage (i), $\Delta U = +37.5 \text{ J}$

$$\Delta U = Q + W$$

$$37.5 = Q + (-25)$$

$$Q = 37.5 + 25 = 62.5 \text{ J}$$

Hence, stage (i) has heat input of 62.5 J.

- SP7 (a) Assuming no heat loss to the surroundings,
 Electrical energy supplied = heat transferred to gas

$$I V t = m c (T_{out} - T_{in})$$

$$0.16 \times 90 = (3.0 \times 10^{-3}) \times 1.8 \times c \times (2.5)$$

$$c = 1070 \text{ J kg}^{-1} \text{ K}^{-1}$$

- (b) Measurements are only taken when the inlet and outlet temperatures have stabilised (i.e. the system has reached steady state). As such, the heat capacity of the apparatus can be ignored.

The heat loss to the surroundings can be eliminated by repeating the experiment again with a different flow rate and adjusting the power of the heating coil, such that the inlet and outlet temperatures remain at the same values as that of the first experiment.

- SP8 (a) In one second,
 Energy supplied = Energy gained

**RAFFLES INSTITUTION
YEAR 56 PHYSICS DEPARTMENT**

$$\begin{aligned}
 IV(1) &= C_{\text{apparatus}}\Delta T + m_{\text{liquid}}c_{\text{liquid}}\Delta T \\
 c_{\text{liquid}} &= \frac{IV - C_{\text{apparatus}}\Delta T}{m_{\text{liquid}}\Delta T} \\
 &= \frac{(3.40)(12.2) - (107)(46.3 \times 10^{-3})}{(0.241)(46.3 \times 10^{-3})} \\
 &= 3273.4 \\
 &= 3.27 \times 10^3 \text{ kg (3 s.f.)}
 \end{aligned}$$

- (b) When the temperature of the liquid is equal to that of the surroundings, the apparatus is neither losing nor gaining heat from the surroundings. There is no need to take into account the heat gain from or loss to the surroundings.

SP9 By the principle of conservation of energy,

Energy lost by steam = Energy gained by ice, water & calorimeter

$$\begin{aligned}
 m_{\text{steam}}l_v + m_{\text{steam}}c_{\text{water}}\Delta T' &= m_{\text{cal}}c_{\text{cal}}\Delta T + m_{\text{water}}c_{\text{water}}\Delta T + m_{\text{ice}}l_f + m_{\text{ice}}c_{\text{water}}\Delta T \\
 m_{\text{steam}}(l_v + c_{\text{water}}\Delta T') &= m_{\text{cal}}c_{\text{cal}}\Delta T + (m_{\text{water}} + m_{\text{ice}})c_{\text{water}}\Delta T + m_{\text{ice}}l_f \\
 m_{\text{steam}}(2.26 \times 10^6 + (4190)(70)) &= (0.120)(380)(30) + (0.250)(4190)(30) + (0.050)(333000) \\
 m_{\text{steam}} &= \frac{(1368 + 31425 + 16650)}{(2.26 \times 10^6 + (4190)(70))} \\
 &= 0.019364 \\
 &= 0.0194 \text{ kg}
 \end{aligned}$$

SP10

$$\begin{aligned}
 Q &= Mc(t_2 - t_1) & \Rightarrow P &= \frac{Mc(t_2 - t_1)}{T_1} \\
 Q &= ml_v & \Rightarrow P &= \frac{ml_v}{T_2}
 \end{aligned}$$

Since power of immersion heater is constant,

$$\begin{aligned}
 \frac{Mc(t_2 - t_1)}{T_1} &= \frac{ml_v}{T_2} \\
 l_v &= \frac{Mc(t_2 - t_1)T_2}{mT_1}
 \end{aligned}$$

Answer: A

SP11

$$\begin{aligned}
 Q &= mc\Delta T & \Rightarrow P &= mc \frac{4.0}{60} \\
 Q &= ml_v & \Rightarrow P &= \frac{ml_v}{40 \times 60}
 \end{aligned}$$

Since energy is supplied at a constant rate, (ie power is constant)

$$\begin{aligned}
 mc \frac{4.0}{60} &= \frac{ml_v}{40 \times 60} \\
 \frac{c}{l_v} &= \frac{1}{160}
 \end{aligned}$$

Answer: D

SP12 Error 1

When water splashes out of the container, there is a less water left in the container. Hence, the amount of heat used to vapourize the water in the container is less than that required for the original mass. The specific latent heat of vapourisation calculated would decrease.

Error 2

When vapour condenses back into the container, there is a greater amount of water in the container. Hence, the amount of heat used to vaporise the water in the container is more than that required for the original mass. The specific latent heat of vaporisation calculated would increase.

Answer: B

SP13 Melting and vaporisation take place at constant temperature

→ Option D is eliminated

Specific latent heat of vaporisation is larger than the specific latent heat of fusion. Hence the time taken for the vaporisation process is much longer than the melting process.

→ Options A and B are eliminated

Specific latent heat of fusion is numerically larger than specific heat capacity. Hence, the time taken for the melting process should be comparable to the time taken for water to be heated from melting point to boiling point.

→ Option C is eliminated

Answer: E